

Aankit Das

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On F1 OPT STEM, Eligible to work in the US without sponsorship | TX

Work History

AI Engineer at Bantrly, Dallas, TX

Mar 2026 to Present

- Built the speech and language systems behind a K-12 learning app by turning student audio into text and scoring it against teaching rubrics
- Built a modular speech-to-text routing engine in Python and FastAPI that separated transcription from LLM scoring across 8 K-12 assessment modules, cutting audio API costs and improving rubric accuracy
- Built a two-layer voice activity detection pipeline combining silence-boundary audio chunking, timestamp deduplication, and a sliding guardrail text window, eliminating false positives during real-time streaming
- Built a provider-agnostic adapter registry across Parakeet TDT, Whisper, AssemblyAI, and Gemini with quality-based fallback and config-driven session health checks, letting engines swap with zero code changes
- Evaluated neural text-to-speech engines pytsx3, Kokoro, edge-tts, Chatterbox, and Chirp against a rubric of intelligibility, naturalness, pacing, and latency per K-12 grade band, producing cost-quality comparisons that guided engine selection
- Designed an LLM-as-judge evaluation pipeline combining structured JSON output, retry logic, and cosine-similarity consistency scoring via nomic-embed-text embeddings across 468 controlled experiments on 3 open-source models, quantifying quality, consistency, and hallucination trade-offs

Machine Learning Researcher at University of Texas at Dallas, Richardson, TX

Jun 2025 to Mar 2026

- Built and optimized neural network pipelines to extract clean signal from noisy space telescope data for an NSF-funded cosmology research project
- Contributed to the CMB-ML open-source research organization, supporting ongoing cosmology and machine learning research toward a planned ICCV or CVPR submission
- Processed and managed 1,600 high-resolution CMB simulated maps totaling 24GB across multiple splits for model training
- Developed a UNet-inspired encoder-decoder neural network architecture trained on modern GPUs in under 20 seconds per epoch to denoise and recover the angular power spectrum from simulated maps
- Ran ablation studies across dropout, batch normalization, loss function variants, and activation functions, comparing MSE, normalized MSE, and Wishart losses alongside ReLU and GELU activations
- Automated hyperparameter sweeps using Hydra and shell scripting to benchmark model performance across ells, detectors, and feature configurations at multiple resolutions

Research Assistant at University of Nebraska-Lincoln, Remote

May 2021 to Jun 2023

- Developed a machine learning segmentation model and benchmark dataset for tracking early plant growth from time-lapse images
- Collaborated with international research teams to develop EmergeNet, a segmentation model for agricultural data that achieved 99.4 percent IoU against a 69 percent baseline for a standard UNet

- Developed the UNL-MED benchmark dataset, curating top-view time-lapse images of maize coleoptiles from pre-emergence through roughly one inch of growth to evaluate EmergeNet
- Applied optimization techniques to monitor maize coleoptile growth, reaching a 0.99 correlation between predicted and observed metrics and publishing the findings in Frontiers and an e-book

Research Assistant at Jadavpur University, Kolkata, WB

Apr 2020 to Aug 2021

- Researched nature-inspired optimization algorithms for audio-based language identification, publishing the results in IEEE journals
- Innovated nature-inspired meta-heuristic feature selection algorithms, BBA-LAHC and HS-NMR, for spoken language identification, achieving over 98 percent test accuracy
- Published two IEEE journal papers documenting advances in NLP and audio signal processing research

Projects

Bantrly Lesson Generator, K-12 ELA Content Generation System

- Built an AI system that generates K-12 speaking lessons by combining a large language model with rule-based checks for safety and quality
- Built a hybrid LLM and rule-based lesson generation system for K-12 ELA speaking practice, combining Groq inference, a CCSS-aligned 80-skill taxonomy, few-shot RAG, and a 9-step validated pipeline to produce structured, grade-appropriate lessons from three inputs
- Designed multi-layer guardrails combining pre-generation input validation, post-generation content checks for vocabulary ceiling and cognitive load, and a cultural bias retry loop that fed flagged terms back into the prompt across up to 5 retries, ensuring no biased lesson was ever saved
- Engineered a defensive validation pipeline for non-deterministic LLM output, auto-correcting three distinct practice field structures across model variants with reflect-stage normalization and domain-specific field stripping to ensure clean JSON regardless of model behavior
- Built a CCSS-aligned skill taxonomy of 80 skills across 4 grade bands and 4 domains with deterministic coverage tracking and cycling, replacing free-form LLM skill generation with controlled, measurable, progression-aware objectives
- Deployed a 6-tab Gradio UI on Hugging Face Spaces combining a student-facing step-by-step lesson experience, a coverage heatmap, a skill taxonomy browser, and a guardrail inspector, giving every view live updates after each generation

Document Intelligence Platform, Production RAG System

- Built a retrieval-augmented generation system that lets users ask questions in plain language and get answers grounded in their own documents
- Engineered a production-grade retrieval-augmented generation system combining local embedding models Ollama and Sentence-Transformers, Chroma DB vector storage, and the Groq llama-3.1-8b-instant LLM API, delivering intelligent document question-answering
- Designed a modular Python architecture of 5 independent modules, chunker, embeddings, vector store, llm, and pipeline, improving testability and maintainability aligned with software engineering best practices

- Built a semantic chunking pipeline with 500-token chunks and 50-token overlap, using PyPDF2 extraction to generate 768-dimensional embeddings for cosine similarity retrieval under 10ms latency on over 100 document chunks
- Demonstrated a cost-optimization strategy, pairing local embeddings for privacy and efficiency with free-tier LLM APIs for fast inference and persistent vector storage for retrieval performance

Education & Certificates

- MS, Computer Engineering, University of Texas at Dallas (GPA - 3.8/4.0) May 2025
- BTech, Electronics and Communication Engineering, University of Calcutta (GPA - 3.7/4.0) Sept 2020

Publications

- **Das, A.** A hybrid meta-heuristic feature selection method for identification of Indian spoken languages from audio signals. IEEE Access.
- **Das, A.** EmergeNet: A novel deep-learning based ensemble segmentation model for emergence timing detection of coleoptile. Frontiers in Plant Science.
- **Das, A.** Hybrid feature selection method based on harmony search and naked mole-rat algorithms for spoken language identification from audio signals. IEEE Access.
- **Das, A.** Flowerphenonet: Automated flower detection from multi-view image sequences using deep neural networks for temporal plant phenotyping analysis. MDPI.